



Hatchetfield Axe-Throwing & Recreation Venues

Campus LAN Segmentation, Trunking & Spanning Tree | Case Study No. 122

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Problem Statement

Hatchetfield's three venues (Downtown, Westside, Harbor District) run on one flat network, with a triangle of links between switches.

Broadcast storms

League nights flood the single broadcast domain and slow business systems.

No isolation

Customers' phones can reach lane-safety sensors that control physical safety.

Layer-2 loop

The triangle needs Spanning Tree to stay loop-free.



Your Solution

VLAN segmentation

VLANs 15, 25, 35, 45, 925 plus unused native VLAN 999

Secure 802.1Q trunks

DTP off, VLANs pruned, native VLAN 999 against VLAN hopping

Router-on-a-stick + ACLs

Inter-VLAN routing; Guest can never reach Lane-Safety

Rapid PVST+

SW-HarborDistrict is root bridge (priority 4096)

LACP EtherChannel

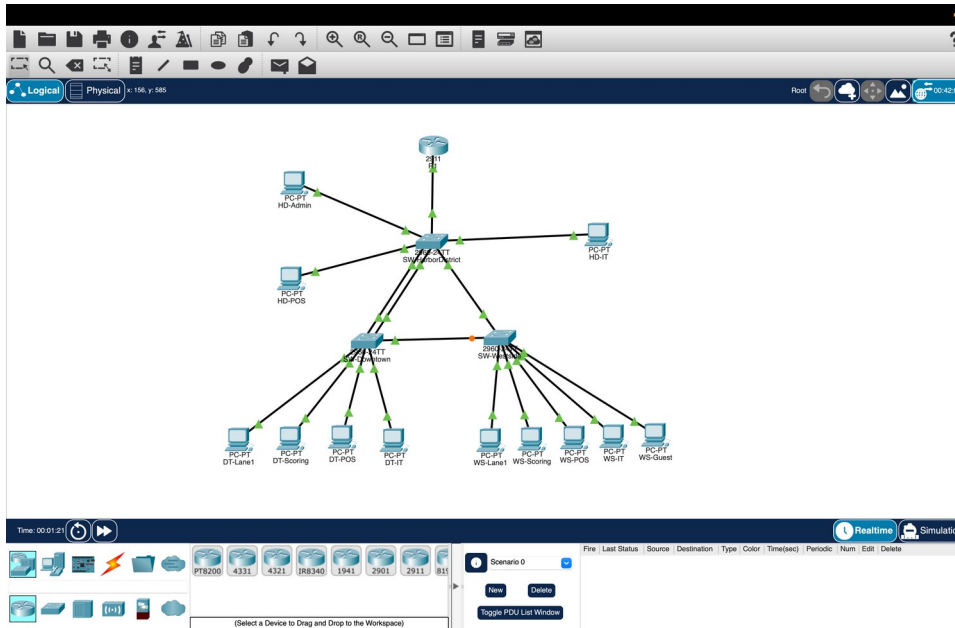
Two Harbor–Downtown links bundled as Po1

Port security + VTP

Max 2 MACs, restrict on Guest ports; VTP transparent



Working (Method / Topology)



1. Design VLANs and assign access ports
2. Trunk all inter-switch links on native VLAN 999
3. Route VLANs on the router; apply ACLs
4. Make HarborDistrict root; bundle links with LACP
5. Verify with show commands and ping tests

Built and verified in Cisco Packet Tracer



Conclusion

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VLANs separate safety, POS, admin, IT and guest traffic

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blocked port keeps the triangle loop-free

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paths from Guest to Lane-Safety-Systems

The redesign removes broadcast storms, isolates guests, and keeps a redundant, loop-free path between all three venues.

Next: Layer-3 switching, DHCP, and Wi-Fi guest portal.





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Thank You